



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	DataCleanAI: Automated Data Quality, Cleaning & Validation Platform	
S. No.	University ID	Name
1	2420030234	Yakkala Guna Veera Sri Amith
2	2420030381	BVS Jayadeep
3	2420030245	K Manoj
Name of the Guide		Ms. Bonagani Prathusha

Abstract

DataCleanAI is an automated data quality, cleaning, and validation platform designed to address the common challenges associated with raw and unstructured datasets. Real-world datasets often contain missing values, duplicate records, inconsistent categorical values, incorrect data types, and numerical outliers, which can negatively affect data analysis and machine learning outcomes. The proposed system allows users to upload datasets in CSV or Excel format and automatically performs data profiling to identify the structure, data types, missing values, unique values, and statistical characteristics of the dataset. It further evaluates the overall quality of the data using metrics such as completeness, uniqueness, validity, and consistency to generate an intuitive data-quality score. Based on the detected issues, the platform applies suitable preprocessing techniques, including median or mode-based imputation, duplicate removal, IQR-based outlier detection and treatment, and standardization of inconsistent text values. Interactive visualizations are provided to help users understand the distribution and quality of their data and compare the dataset before and after cleaning. The system is developed using Python, Pandas, NumPy, Scikit-learn, Plotly, and Streamlit, providing a simple and interactive interface that can be used without complex infrastructure. The cleaned dataset can subsequently be downloaded and used for further statistical analysis, visualization, or machine learning applications. Overall, DataCleanAI aims to reduce the manual effort involved in data preprocessing and provide an efficient, reusable, and user-friendly solution for improving the quality and reliability of structured datasets.

